

1.

- A. Y_i is the i -th student's GPA at the end of their freshman year. X_i is the i -th student's ACT test score. n is the number of observed freshman students in the study, which is given to be 120 students.
- B. β_0 is the intercept parameter for the least squares regression line. 0 is outside the range of observed X 's, so it should not be interpreted in context. β_1 is the slope parameter of the least squares regression line. For every one-point increase in a freshman student's ACT test score, the average change in GPA is expected to be β_1 .
- C. The aspect of the model that the three data points illustrate is the error term for each observation. The three students have the same X value but different Y values, which the error term accounts for.
- D. The linear regression model does not appear to be appropriate based on the appearance of the scatterplot. The scatterplot does not show a strong apparent linear pattern and the correlation coefficient $r = 0.269$ indicates a weak association. There is not also much of an ellipsoidal shape which would have been ideal for linear regression.

2.

- A. $\hat{Y} = 2.11405 + 0.03883(26) = 2.11405 + 1.00958 = 3.12363$. The predicted freshman GPA for a student with ACT test score $X = 26$ is 3.12363.
- B. The estimated difference in the mean GPA when the ACT score differs by three points is $3(0.03883) = 0.11649$.

3. $H_0: \beta_1 = 0$ vs. $H_a: \beta_1 \neq 0$ at $\alpha = 0.01$

$$t_{\text{obs}} = \frac{\widehat{\beta_1} - \beta_1}{\widehat{SD}(\beta_1)} = \frac{0.03883}{0.01277} = 3.04$$

null distribution: $t_{n-2} = t_{118}$

p-value = 0.00292

With a p-value of 0.00292, we reject H_0 at $\alpha = .01$. There is very strong evidence that there is an association between student's ACT test scores and GPA at the end of their freshman year.

The director of admissions of the college would care about whether the hypothesis test result is statistically significant because it could be an indicator of how well the student would do in their classes. If ACT score is a good predictor, they could use it to determine

acceptance. If ACT score is not a good predictor, they would probably rely on other factors to determine acceptance.

4. The formula for standard error of β_1 is $\frac{S_Y\sqrt{1-r^2}}{S_X\sqrt{n-1}}$, given the summary statistics. S_Y is given as GPA SD = 0.644. r is given in the summary statistics to be 0.269. S_X is provided in the summary statistics as ACT SD = 4.472. 120 students are in the study so $n-1=119$. Then $\frac{0.644\sqrt{1-0.269^2}}{4.472\sqrt{119}} = 0.01271$. This is because $S_Y\sqrt{1-r^2}$ is an estimate for $\hat{\sigma}$ and not exact. $\hat{\sigma}$ however, is provided in the R output as “Residual standard error: 0.6231” and would be substituted for the numerator.
5. The director is incorrect in the interpretation of the R output. The director’s explanation of an individual future student’s predicted GPA from their ACT score would require calculations from a prediction interval since would be for a single observation and has more variability than the average of several observations. The R output produced results for a confidence interval which should be interpreted as “We are 95% confident that future students admitted with an ACT score of 26 will have an average freshman year GPA between B (3.006) and B+ (3.24).”
6. From the ANOVA table, the formula for R^2 is $R^2 = \frac{SSR}{SSTO} \rightarrow R^2 = \frac{3.588}{3.588+45.818} = 0.0726$. R^2 can also be calculated by squaring the given $r = 0.269$ or by looking at the “Multiple R-squared” value which are both 0.0726 as well. The coefficient of determination represents the proportionate drop in variation of Y , due to regression on X . Then 7.26% of the variation in freshman year GPAs is explained by the regression on ACT test scores. From this information, there is a weak association between ACT test scores and freshman year GPAs. R^2 is close to 0, so the least squares line is almost horizontal.